TAO LPDNet and LPRNet license plate detection and recognition

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> Applicable environment: Jetson Orin NX, JetPack 7.2.1, DeepStream 9.1.

1. Case goal and processing flow

TrafficCamNet detects the vehicles first, LPDNet locates the region of the license plate, and LPRNet recognizes the characters on the cropped license plate. The Chinese version and the US version use different models and different dictionaries, so they have to be selected according to the region.

```
Video/USB/CSI → TrafficCamNet → LPDNet license plate detection → LPRNet character
recognition
             → the preview/the video output
```

2. The checks before running it

```
cd ~/deepstream_tao_apps/deepstream_app_tao_configs

# Confirm the configuration and the input source
grep -E "uri=|config-file=" run_lpr_china.txt
grep -E "uri=|config-file=" run_lpr_us.txt

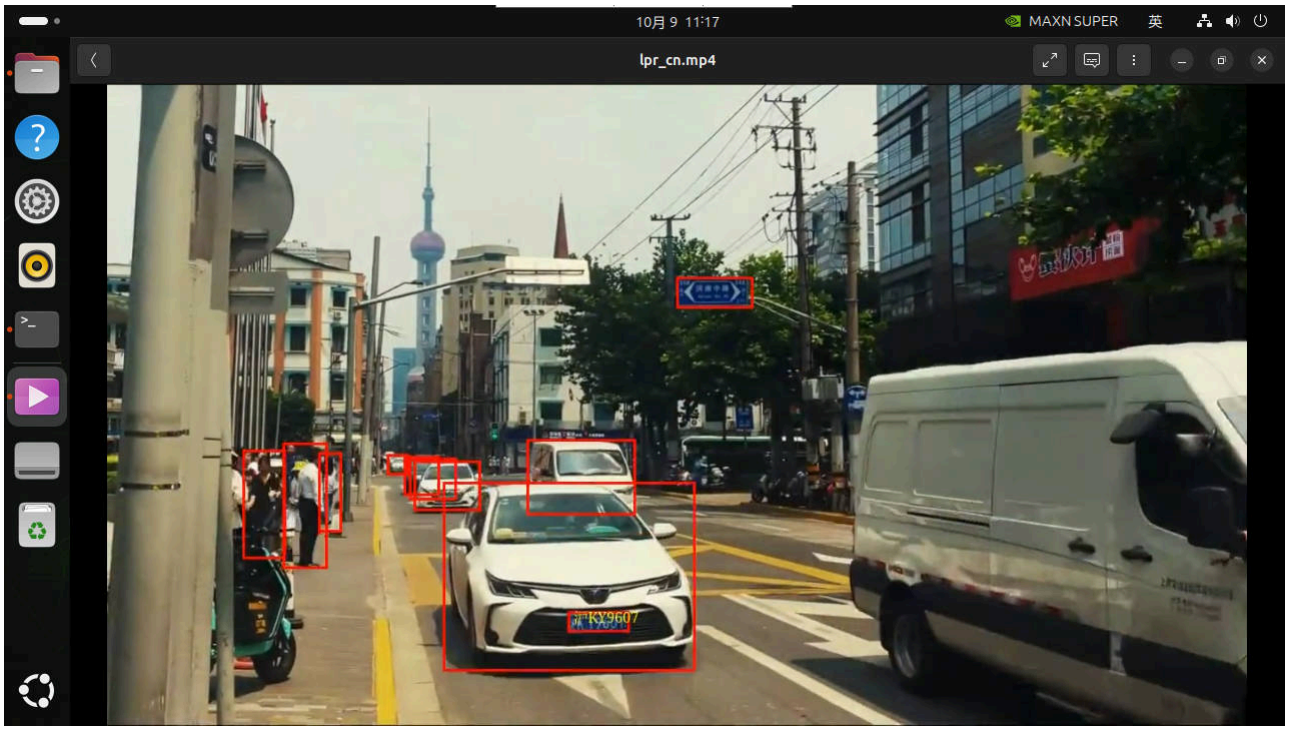
# Confirm the models and the dictionaries
ls -la /opt/nvidia/deepstream/deepstream-
9.1/samples/models/tao_pretrained/lprnet_vdeployable_onnx_v1.1/
ls -la /opt/nvidia/deepstream/deepstream-
9.1/samples/models/tao_pretrained/lpdnet_vpruned_v2.3.1/
ls -la ~/deepstream_tao_apps/apps/tao_others/deepstream_lpr_app/dict_ch.txt
ls -la ~/deepstream_tao_apps/apps/tao_others/deepstream_lpr_app/dict_us.txt
```

Confirm that the material of the target region exists. The real-time camera uses the Jetson desktop/VNC; the camera image should have clear and sufficiently large vehicles and license plates.

3. Start the detection

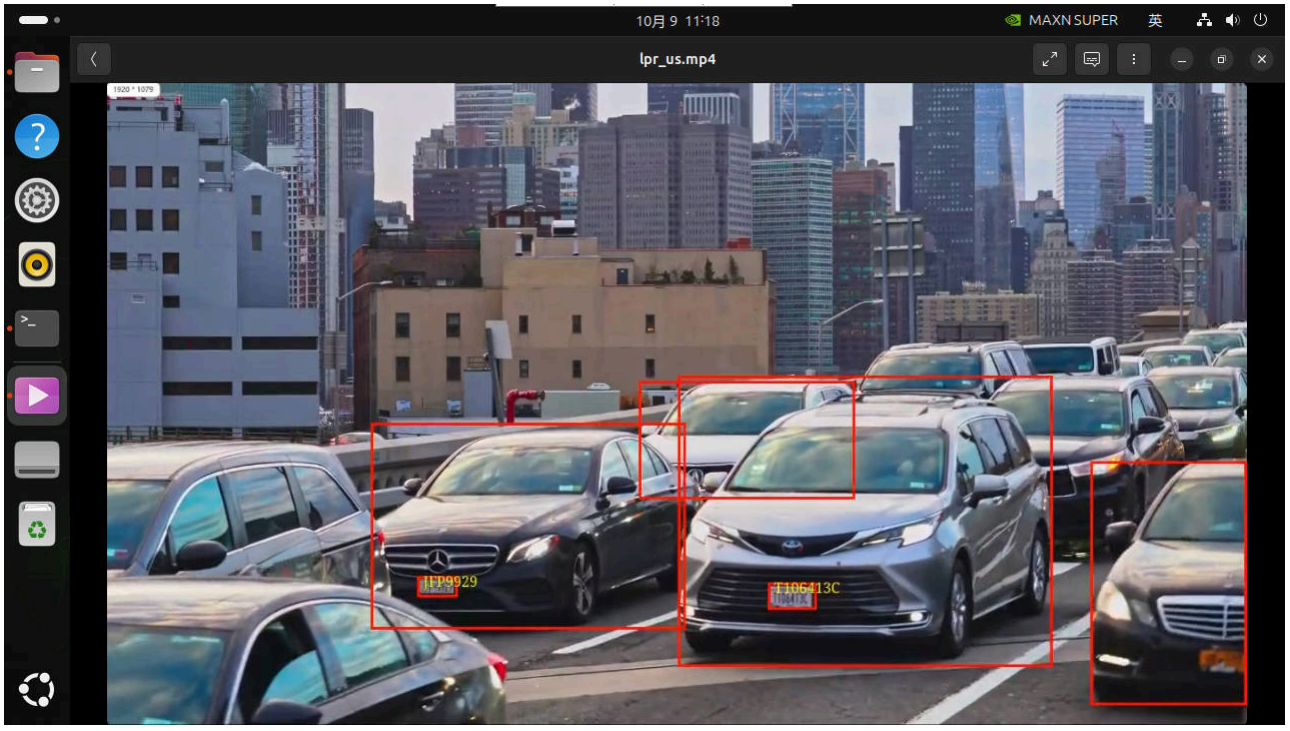
The Chinese version (the video):

```
cd ~/deepstream_tao_demo && export DISPLAY=:0 && ./run_lpr_cn.sh
```



The US version (the video):

```
cd ~/deepstream_tao_demo && export DISPLAY=:0 && ./run_lpr_us.sh /absolute/path/us.mp4
```



The corresponding inputs are:

- The Chinese version: `~/Videos/Video007.mp4`
- The US version: `~/Videos/us.mp4`; if it does not exist, the script uses `02.DeepStream development/02.License plate detection and recognition/Test cases/us.mp4`

The corresponding configuration files:

- The Chinese version: `~/deepstream_tao_apps/deepstream_app_tao_configs/run_lpr_china.txt`
- The US version: `~/deepstream_tao_apps/deepstream_app_tao_configs/run_lpr_us.txt`

The license plate boxes and the character text represent the result of the localization stage and of the recognition stage respectively.

To replace the test video: pass the absolute path directly to `run_lpr_cn.sh` or `run_lpr_us.sh`; the script generates a temporary configuration and it does not rewrite the template in the repository any more. You must not only replace the material while still using the model/dictionary of the other region.

4. The real-time detection with a USB camera

The LPR USB entry uses the parameter of the specified region. First check the USB camera node and the formats:

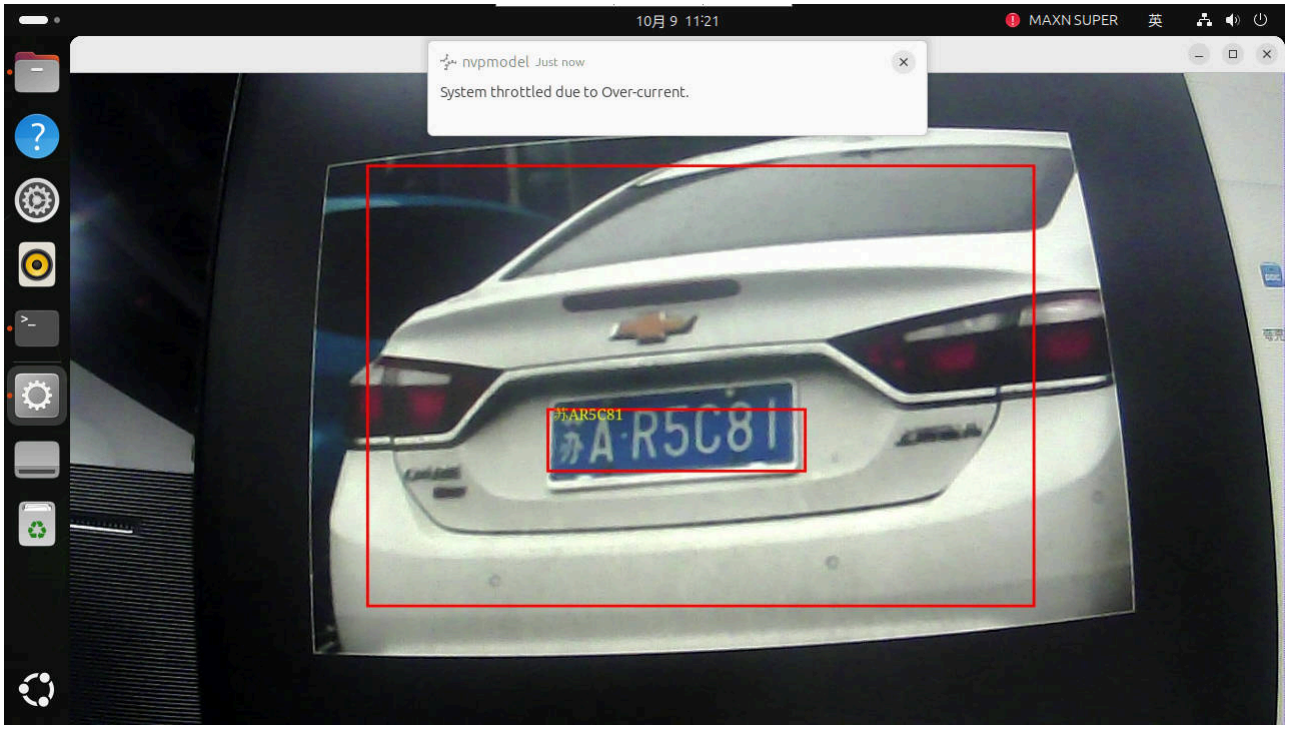
```
ls -l /dev/video*
v4l2-ctl -d /dev/video0 --list-formats-ext
```

The Chinese version:

```
cd ~/deepstream_tao_demo && export DISPLAY=:0 && ./run_case.sh lpr_china usb
```

The US version:

```
cd ~/deepstream_tao_demo && export DISPLAY=:0 && ./run_case.sh lpr_us usb
```



The real-time preview does not produce an offline MP4; the recognition result depends on the resolution of the camera, on the shutter/motion blur and on the pixel size of the license plate in the frame.

Press `Ctrl+C` once and wait for the application to exit and for the camera to be released.

5. The real-time detection with a CSI camera

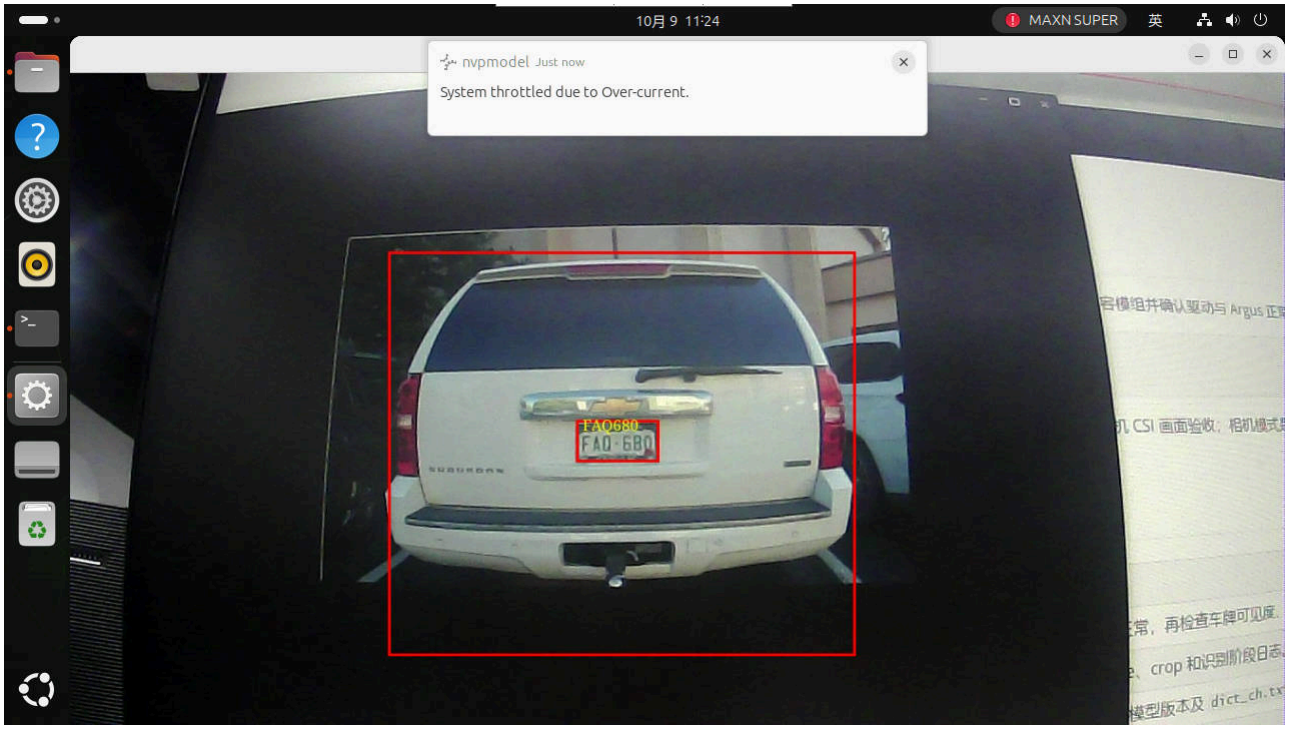
The CSI entry supports the real-time preview; it uses the Jetson Argus instead of the USB V4L2 device path. After connecting a compatible module and confirming that the driver and the Argus work normally, run:

The Chinese version:

```
cd ~/deepstream_tao_demo && export DISPLAY=:0 && ./run_case.sh lpr_china csi
```

The US version:

```
cd ~/deepstream_tao_demo && export DISPLAY=:0 && ./run_case.sh lpr_us csi
```



CSI uses the native Argus `type=5` and it does not read `/dev/video10`. The entry supports the real-time preview and it does not produce an MP4.

6. Frequently asked questions

Symptom	The check and the handling
There are no license plate boxes	First confirm that the vehicle detection boxes are normal, and then check the visibility and the size of the license plate and the LPDNet configuration. **The Chinese and the US plates must not use the wrong LPD**: the Chinese one uses `sgie_lpd_DetectNet2_ch.txt` and the US one uses `sgie_lpd_DetectNet2_us.txt`.
There are license plate boxes but no text	Check the LPRNet engine, the crop and the log of the recognition stage.

Symptom	The check and the handling
The characters are garbled/the region does not match	Check the run script, the model version and `dict_ch.txt` / `dict_us.txt`.
`open dictionary file failed` + `std::bad_alloc`	The current working directory has no `dict.txt`, or the symlink points to a file that does not exist. The six scripts already have the dictionary switching logic built in, so you can use `run_lpr_*.sh` directly.
The US plate shows garbled Chinese	Confirm that you use `run_lpr_us*.sh`; the script switches to `LPDNet_usa + us_lprnet + dict_us`.
There is no image with USB	Check the node, the format, the permissions, the occupancy and the display session. The resolution has to match the `camera-width/camera-height/camera-fps` in the configuration file (640x480@15 in the YUYV mode).
The CSI did not start	Do not switch manually to the writer; the script stops the relay internally and opens the Argus directly. Check the module, the sensor driver, `nvargus-daemon` and whether the camera is in use.
`nvv4l2h264enc has no property named 'gpu-id'`	It is a known non-blocking warning of DS 9.1 and it does not affect the inference or the display.

7. References

- The NVIDIA DeepStream TAO application examples: https://github.com/NVIDIA/DeepStream/tree/main/src/apps/tao_apps
- The DeepStream TAO integration documentation: https://docs.nvidia.com/metropolis/deepstream/dev-guide/text/DS_TAO_integration.html